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Adjustable Tensioning Spool for Steerable Catheters

Abstract

A catheter system may include a steering catheter and a steering cable extending through a lumen in the wall of the steering catheter. A steering assembly may include a spool assembly having an outer spool, an inner spool received within the outer spool, and a spool shaft received within the inner spool. The spool shaft and a recess of the inner spool may be shaped so that rotation of the spool shaft causes rotation of the inner spool. The inner spool may include splines that engage with complementary recesses of the inner spool, such that the outer spool is prevented from rotating relative to the inner spool. The steering cable may be fixed to the outer spool and may be configured to be wound around the spool assembly upon rotation of the outer spool about the longitudinal axis of the spool shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to the filing date of U.S. Provisional Patent Application No. 63/563,562, filed Mar. 11, 2024, the disclosure of which is hereby incorporated by reference herein.

BACKGROUND OF DISCLOSURE

[0002] The present disclosure relates generally to devices, systems and methods for delivering an interventional device into a patient for implantation.

[0003] Intravascular medical procedures allow for the performance of therapeutic treatments in a variety of locations within a patient's body while requiring only relatively small access incisions. An intravascular procedure may, for example, eliminate the need for open-heart surgery, reducing risks, costs, and time associated with an open-heart procedure. The intravascular procedure may also enable faster recovery times with lower associated costs and risks of complications. An example of an intravascular procedure that reduces procedure and recovery time and cost over conventional open heart surgery is a heart valve replacement or repair procedure in which an artificial valve or valve repair device is guided to the heart through the patient's vasculature. For example, a catheter is inserted into the patient's vasculature and directed to the inferior vena cava. The catheter is then urged through the inferior vena cava toward the heart by applying force longitudinally (e.g., pushed forward) to the catheter. Upon entering the heart from the inferior vena cava, the catheter enters the right atrium. The catheter may be guided across the atrial septum (e.g., via a guidewire that has already been passed through the atrial septum) into the left atrium. The distal end of the catheter may be deflected by one or more deflecting mechanisms in order to align the distal end of the catheter, as well as a medical device positioned therein, with the mitral valve. Catheter deflection can be achieved by tension cables, or other mechanisms positioned inside the catheter. Precise control of the distal end of the catheter allows for more reliable and faster positioning of a medical device and/or implant and other improvements in the procedures.

BRIEF SUMMARY

[0004] According to one aspect of the disclosure, a catheter system includes a steering catheter having a wall and a first lumen formed within the wall. A first steering cable may extend through the first lumen along a length of the wall of the steering catheter toward a distal end portion of the steering catheter, the first steering cable being coupled to the distal end portion of the steering catheter. A steering assembly may include a first spool assembly, the first spool assembly including a first outer spool, a first inner spool received at least partially within a recess of the first outer spool, and a spool shaft having a first end portion received at least partially within a recess of the first inner spool. The first end portion of the spool shaft and the recess of the first inner spool may each be shaped so that rotation of the spool shaft about a longitudinal axis thereof transmits torque to the first inner spool to cause rotation of the first inner spool about the longitudinal axis of the spool shaft. The first inner spool may include a spline section forming a plurality of splines, and the recess of the first outer spool may have a shape that is complementary to the plurality of splines, such that when the first inner spool is received at least partially within the recess of the first outer spool, the plurality of splines engage the complementary shape of the recess of the first outer spool to prevent rotation of the first inner spool relative to the first outer spool about the longitudinal axis of the spool shaft. The first steering cable may be fixed to the first outer spool and configured to be wound around the first spool assembly upon rotation of the first outer spool about the longitudinal

axis. The first inner spool may be configured to be received within the first outer spool in any one of a plurality of unique relative rotational positions, so that the first steering cable is configured to have a first amount of pretension in a first one of the plurality of unique relative rotational positions and a second amount of pretension in a second one of the plurality of unique relative rotational positions, the first amount of pretension being different than the second amount of pretension. The steering assembly may include a base, the spool shaft may have a central portion received within the base, and the first spool assembly may be positioned on a first surface of the base. The steering assembly may include a second spool assembly positioned on a second surface of the base opposite the first surface of the base. The second spool assembly may include a second outer spool, and a second inner spool received at least partially within a recess of the second outer spool. The first inner spool may be identical to the second inner spool, and the first outer spool may be identical to the second outer spool. The spool shaft may have a second end portion received at least partially within a recess of the second inner spool, and the second end portion of the spool shaft and the recess of the second inner spool may each be shaped so that rotation of the spool shaft about the longitudinal axis thereof transmits torque to the second inner spool to cause rotation of the second inner spool about the longitudinal axis of the spool shaft. The second inner spool may include a spline section forming a plurality of splines, and the recess of the second outer spool may have a shape that is complementary to the plurality of splines of the spline section of the second inner spool, such that when the second inner spool is received at least partially within the recess of the second outer spool, the plurality of splines of the spline section of the second inner spool engage the complementary shape of the recess of the second outer spool to prevent rotation of the second inner spool relative to the second outer spool about the longitudinal axis of the spool shaft.

[0005] The catheter system may further include a second lumen formed within the wall of the steering catheter, a second steering cable extending through the second lumen along a length of the wall of the steering catheter toward the distal end portion of the steering catheter, the second steering cable being coupled to the distal end portion of the steering catheter. The second steering cable may be fixed to the second outer spool and configured to be wound around the second spool assembly upon rotation of the second outer spool about the longitudinal axis. The first steering cable may be fixed to the first outer spool and the second steering cable may be fixed to the second outer spool such that, when the spool shaft is rotated about the longitudinal axis to cause simultaneous rotation of the first outer spool and the second outer spool, the first steering cable winds around the first spool assembly while the second steering cable unwinds from the second spool assembly. The first lumen may be positioned on a generally opposite side of the steering catheter compared to the second lumen. Tensioning the first steering cable may be configured to deflect the distal end portion of the steering catheter in a first steering direction, and tensioning the second steering cable may be configured to deflect the distal end portion of the steering catheter in a second steering direction, the first steering direction being generally opposite the second steering direction. The first lumen may be positioned diametrically opposite to the second lumen.

Tensioning the first steering cable may be configured to deflect the distal end portion of the steering catheter in a first steering direction, and tensioning the second steering cable may be configured to deflect the distal end portion of the steering catheter in a second steering direction, the first steering direction and the second steering direction being positioned in a same plane but in opposite directions along the same plane. The system may further include a drive gear, the drive gear having a central opening, the second end portion of the spool shaft being received within the central opening. The second end portion of the spool shaft and the central opening recess of the drive gear may each be shaped so that rotation of the drive gear about the longitudinal axis of the spool shaft transmits torque to the spool shaft to cause rotation of the spool shaft about the longitudinal axis of the spool shaft.

[0006] The steering assembly may further include an actuator shaft having a first end portion and a second end portion, the second end portion being received within and coupled to the base. The

steering assembly may further include a gear wheel having a central opening, the first end portion of the actuator shaft being received within the central opening of the gear wheel such that rotation of the actuator shaft about a longitudinal axis thereof causes rotation of the gear wheel about the longitudinal axis of the actuator shaft. The gear wheel may be operably coupled to the drive gear such that rotation of the gear wheel transmits torque to the drive gear. The steering assembly may be received within a handle of a delivery system, and the actuator shaft may have a terminal end protruding outside of the handle. A steering knob may be coupled to the terminal end of the actuator shaft such that rotation of the steering knob causes corresponding rotation of the actuator shaft. The steering assembly may include a first hard stop operably coupled to the drive gear, the first hard stop configured to prevent rotation of the drive gear beyond a first threshold amount of rotation in a first rotational direction. The steering assembly may include a second hard stop operably coupled to the drive gear, the second hard stop configured to prevent rotation of the drive gear beyond a second threshold amount of rotation in a second rotational direction opposite the first rotational direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a perspective view of a delivery system configured for delivering, positioning, and deploying a prosthetic heart valve, including a handle assembly and a catheter assembly.

[0008] FIG. 2 is an exploded view of a portion of the handle assembly of the delivery system of FIG. 1, including valve cover retraction knob and cone assembly.

[0009] FIG. 3 is a side view of a portion of the catheter assembly of the delivery system of FIG. 1, including a catheter shaft, coiled and/or braided section, valve cover, and nosecone.

[0010] FIG. 4A is a transverse cross-sectional view of the catheter shaft of FIG. 3, showing various components thereof.

[0011] FIGS. 4B-4C are transverse cross-sectional views of a catheter assembly, operable with the delivery system of FIG. 1, according to another aspect of the disclosure.

[0012] FIG. 5A is a longitudinal cross-section of the catheter assembly of FIG. 3, showing the nested arrangement of various components thereof.

[0013] FIG. 5B is a longitudinal cross-section of a catheter assembly, operable with the delivery system of FIG. 1 and according to another aspect of the disclosure, showing the nested arrangement of various components thereof.

[0014] FIG. 6 is an isolated view of a suture rigging assembly that may be coupled to a suture catheter of any of the delivery systems described herein.

[0015] FIG. 7 is a perspective view of the suture rigging assembly connecting the suture catheter to the prosthetic heart valve.

[0016] FIGS. 8-9 are top perspective views of portions of a steering assembly according to an aspect of the disclosure.

[0017] FIG. 10 is a cross-section of the portion of the steering assembly of FIGS. 8-9.

[0018] FIG. 11 is a perspective view of a spool shaft of the steering assembly of FIGS. 8-10.

[0019] FIG. 12 is a perspective view of an outer spool of the steering assembly of FIGS. 8-10, shown in partial phantom.

[0020] FIG. 13 is a perspective view of an inner spool of the steering assembly of FIGS. 8-10.

DETAILED DESCRIPTION

[0021] As used herein, the term “inflow end,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve through which blood enters when the heart valve is functioning as intended, whereas the term “outflow end,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve through which blood exits when the heart

valve is functioning as intended. For a prosthetic mitral valve, the inflow end is closest to the left atrium when the heart valve is implanted in a patient, and the outflow end is closest to the left ventricle when the heart valve is implanted in a patient. Further, when used herein in connection with a delivery device, the terms “proximal” and “distal” are to be taken as relative to a user operating the device in an intended manner. “Proximal” is to be understood as relatively close to the user and “distal” is to be understood as relatively farther away from the user. Also as used herein, the terms “substantially,” “generally,” and “about” are intended to mean that slight deviations from absolute are included within the scope of the term so modified.

[0022] FIG. 1 is a perspective view of an example of a delivery system **100** configured for delivering, positioning and deploying a prosthetic heart valve, including a handle assembly **105** and a catheter assembly **150**. FIG. 2 is an exploded view of a portion of the handle assembly **105** of the delivery system **100** of FIG. 1, including valve cover retraction knob **115** and cone assembly **120** (which may also be referred to as a locking assembly herein). FIG. 3 is a side view of a distal end portion of the catheter assembly **150** of the delivery system **100** of FIG. 1, including a catheter shaft **155**, coiled and/or braided section **160**, valve cover **165** (which may also be referred to as a valve capsule), and an atraumatic distal tip or nosecone **170**.

[0023] Delivery system **100** generally includes a handle assembly **105** and a catheter assembly **150**. Catheter assembly **150** extends from a proximal end coupled to handle assembly **105** to an atraumatic tip or nosecone **170** (not shown in FIG. 1) at a distal end and includes a plurality of catheter and/or hypotube components, at least some of which are longitudinally slidable relative to one another and provide different functionality during operation of delivery system **100** to enable effective delivery and deployment of a prosthetic heart valve, such as a prosthetic mitral valve. While the catheter assembly **150** is depicted as having a length relative to handle assembly **105**, it should be understood that the catheter assembly **150** can have any suitable length and, in some examples, has a much greater length than the handle assembly **105**.

[0024] Attached to a housing **105a** of the handle assembly **105** can be various control mechanisms for controlling one or more aspects of the delivery. For example, attached to the housing **105a** of the handle assembly **105** can be one or more knobs **110** for controlling steering of the of the catheter assembly **150**. Additional details regarding steering operations that may be utilized in conjunction with the components and features described herein are described in U.S. Patent Application Publication No. 2023/0364387, the disclosure of which is hereby incorporated by reference herein. The handle assembly **105** can also include a valve cover retraction knob **115** and a locking or cone assembly **120**. The valve cover retraction knob **115** can be rotated to allow for retraction of valve cover **165** during delivery and/or expression of a prosthetic heart valve. In some examples, rotating valve cover retraction knob **115** results in proximal translation of a catheter member that is attached to the valve cover **165**, allowing for the valve cover **165** to uncover a self-expandable prosthetic heart valve to allow the prosthetic heart valve to self-expand. The cone assembly **120** can be rotated relative to housing **105a** to lock or unlock an outer catheter relative to an inner catheter, although the functionality of the cone assembly **120** is not described in greater detail herein. In some embodiments, the cone assembly **120** may be omitted.

[0025] The catheter assembly **150** can include a catheter shaft **155**, a coiled and/or braided section **160**, a valve cover **165**, and a nosecone **170** (which is not shown in FIG. 1), as will be described in greater detail below. It should also be understood that catheter assembly **150** may include a number of additional catheter members nested within catheter shaft **155**, as described in greater detail below.

[0026] FIG. 4A is a transverse cross-sectional view of the catheter shaft **155** of FIG. 3, showing various components thereof, including addition catheter components nested therein.

[0027] As shown in FIG. 4A, which is a transverse cross-sectional view of catheter assembly **150** taken along line 4-4 of FIG. 3, these components may include an outer sheath **405a** (which may include the catheter shaft **155**, coiled and/or braided section **160**, and/or valve cover **165**), a

steering catheter **410a**, an extension catheter **415a**, a suture catheter **420a**, and a nosecone catheter **425a**, all arranged in a concentric nested relationship. The arrangement of these components, as well as valve cover **550a**, nosecone **575a**, and valve retainer **590a** (also referred to as a “can”) is shown in the longitudinal cross-section of catheter assembly **150** shown in FIG. 5A. During delivery, the proximal petals (e.g. the proximal apices of a frame, such as an outer frame) of the compressed prosthetic valve can be nested inside of the valve retainer **590** when the prosthetic valve is held back by the suture catheter, which is under tension. In one example, valve retainer **590a** may be attached directly to a tip ring of the steering catheter **410a**. As illustrated in the figures, nosecone catheter **425a** has a lumen sized to receive a guidewire **430a** therein. Each of these components is described in detail below. It should be understood that the valve cover **165** of FIG. 1 and FIG. 3 may be substantially the same as or identical to the other valve covers described herein. Similarly, it should be understood that the atraumatic distal tip or nosecone **170** of FIG. 3 may be substantially the same as or identical to the other atraumatic distal tips or nosecones described herein. In some examples, the suture catheter **420a** may have a plurality of sutures or other wire loops attached to a distal end thereof, the suture loops or wire loops configured to selectively maintain a connection between the prosthetic heart valve and the delivery device.

[0028] To selectively control the curvature of a distal section of steering catheter **410a**, the steering catheter **410a** may be provided with a plurality of tension cables (not shown). The tension cables may travel from handle assembly **105** (e.g., one or more of the knobs **110**) through a plurality of lumens **412a** (which in some embodiments may be polymer tubes, such as Nylon, Pebax, polyimide, or polytetrafluoroethylene or “PTFE” tubes, or any other type of polymer) to the distal end of steering catheter **410a**. In one embodiment, steering catheter **410a** may include four lumens **412a** equally spaced at 90° intervals around the circumference of steering catheter **410a**. In the embodiment of FIG. 4A, steering catheter **410a** may include four pairs of lumens **412a** (a total of eight lumens **412a**), with the pairs of lumens equally spaced at 90° intervals around the circumference of steering catheter **410a**. Any number of these lumens **412a** (or pairs of lumens) may be provided depending on the various directions of deflection that may be desired. For example, six pairs of lumens **412a** may be provided, with each pair of lumens **412a** housing one steering cable of a pair of steering cables coupled to a corresponding steering knob **110** (e.g., two steering cables attached to each steering knob **110**, with each steering cable extending through one pair of lumens **412a**, for a total of twelve lumens **412a** arranged in six pairs).

[0029] FIGS. 4B-4C are transverse cross-sectional views of a catheter assembly **150b**, operable with the delivery system of FIG. 1 (e.g., in place of catheter assembly **150**), according to another aspect of the disclosure.

[0030] In this example, catheter assembly **150b** omits an extension catheter (e.g., extension catheter **415a** described above) and includes an outer catheter **405b** (which may include the catheter shaft, coiled and/or braided section, and/or valve cover similar to those shown and described in connection with FIG. 3), a steering catheter **410b**, an intermediate catheter **415b**, a suture catheter **420b**, and a nosecone catheter **425b**. Also in this example, the steering catheter **410b** includes six pairs (a total of 12) of lumens **412b**, which again may be formed with polymer tubes.

[0031] The arrangement of these components, as well as valve cover **550b**, nosecone **575b**, and valve retainer **590b** is shown in the longitudinal cross-section of catheter assembly **150b** shown in FIG. 5B.

[0032] A suture rigging assembly **600** that assists in collapsing and drawing prosthetic heart valve V into delivery device **100** is shown in FIGS. 6-7. Suture rigging assembly **600** may also help keep the prosthetic heart valve V connected to the delivery device **100**, as well as helping control the deployment of the prosthetic heart valve V during use. Suture rigging assembly **600** may include a coupling ring **601** to which a plurality of suture tethers may be attached. An exemplary suture rigging assembly and coupling ring is shown in U.S. Pre-Grant Publication 2023/0030110, the disclosure of which is hereby incorporated by reference.

[0033] One or more suture threads **605** may be attached to the head of coupling ring **601**. The suture threads **605** may be comprised of various materials, both man-made and natural. Examples of natural suture materials may include, but are not limited to, silk, linen, and catgut. Examples of synthetic suture materials may include, but are not limited to, textiles such as nylon or polyester, or flexible metallic cables. An elongated suture thread **605** may be threaded through a plurality of the bores **610** in coupling ring **601** to form tethers **615**. Suture rigging assembly **600** may include a coupling ring **601** having at least one tether **615** or a plurality of tethers. For example, suture thread **605** may be threaded distally through a bore **610** in inner ring of coupling ring **601** and then proximally through an adjacent bore in outer ring of coupling ring **601**, thereby forming an elongated loop or tether **615** extending distally from coupling ring **601**. Thus, tether **605** includes two lengths of suture thread extending side-by-side and continuous with one another at their distal ends. Suture thread **605** may then be threaded distally through an adjacent bore **610** in inner ring and then proximally through an adjacent bore in outer ring, thereby forming another elongated loop or tether **615** extending distally from coupling ring **601**. This pattern may be repeated to form a plurality of elongated loops or tethers **615** around the entire circumference of coupling ring **601**. Thus, tethers **615** may be formed by a single continuous suture thread **605**, with the leading and trailing ends of the suture thread being joined to one another by one or more terminating knots.

[0034] Suture threads forming a tether **615** may be joined together by a first knot or stop knot **620** at a spaced distance from coupling ring **601**. Stop knots **620** reduce the ability of suture threads to separate too far from one another or to create a large loop or lasso. The distance between the knots on a tether **615** will define the maximum loop or lasso that can be formed by the tether. As a result of using knots, any loop or lasso able to form will be smaller in size than the loop or lasso in a tether **615** that does not have any knots. Preventing the formation of large loops or lassos is important because a large loop or lasso may become entangled with the apexes of the ventricular anchor of a prosthetic heart valve, thereby impairing the user's ability to pull back the entangled tether **615** after valve has been deployed. As shown in FIG. 6, stop knots **620** in adjacent tethers **615** preferably are formed at different spaced distances from coupling ring **601**. For example, the stop knots **620** in a first group of tethers **615** may be spaced a first distance from coupling ring **601**, and the stop knots in a second group of tethers that alternate with the tethers in the first group may be spaced a second distance greater than the first distance from the coupling ring. By offsetting the stop knots **620** in adjacent tethers **615** from one another, the tethers are better able to collapse to a smaller, more compact cross-sectional size within the confines of delivery device **100**.

[0035] The stop knots **620** in tethers **615** create in each tether an upper or proximal connecting loop **625** between the knot and coupling ring **601**. The ability of suture thread **605** to move freely within bores **610** enables the lengths of tethers **615** to self-adjust to a certain degree. That is, each tether **615** is free to move proximally until its stop knot **620** contacts coupling ring **601** and is free to move distally until the stop knots in the adjacent tethers contact the coupling ring. Therefore, as one tether **615** lengthens as it moves distally, there is a corresponding proximal movement and shortening of the adjacent tethers on either side of it, in the manner of a pulley. This adjustment in the lengths of tethers **615** enables a balancing of the load imparted to each of the tethers as prosthetic heart valve is collapsed during loading into delivery device **100** or during re-sheathing. For example, if a shorter tether **615** experiences a higher tensile stress upon the loading of prosthetic heart valve into delivery device **100**, that tether may lengthen as the adjacent tethers shorten until the tensile stress on all of the tethers reaches an equilibrium point at which the total tensile stress is substantially evenly distributed among all of the tethers. Maintaining a balanced load among tethers **615** prevents any one of the tethers from becoming overloaded and breaking, which can impede the functionality of the entire system. Further, more evenly distributing the load among tethers **615** enables the overall tensile capacity of suture rigging assembly **600** to be increased.

[0036] Additional knots may also be formed at the distal or closed end of tethers **615**. As shown in

FIG. 6, a third knot or lower fixture knot **630** may be formed at a spaced distance from the distal end of tether **615**, forming a closed attachment loop **635** therein. Attachment loops **635** are intended to hook onto the pins of prosthetic heart valve and to apply tension to assist in collapsing the prosthetic heart valve during loading into delivery device **100**, as described more fully below. Preferably, attachment loops **635** have a relatively small size so that, following their release from pins during deployment, they do not become entangled with the pins or other structures of prosthetic heart valve, impeding proper deployment of the heart valve and removal of delivery device **100** from the patient.

[0037] To help visualize the locations of tethers **615**, and in particular the positions of attachment loops **635**, during the deployment of prosthetic heart valve in a patient, some embodiments of suture rigging assembly **600** may include a radiopaque marker **640** on all or at least some of the tethers. Radiopaque markers **640** may be formed of any material that can be readily visualized under fluoroscopy, including metals such as gold, platinum, platinum-iridium, tantalum, tantalum-tungsten, and others, and may take any shape. Preferably, radiopaque markers **640** have a bore or channel extending therethrough so that the markers may be threaded onto suture threads before lower fixture knot **630** is formed therein or as suture thread **605** is threaded through bores **610**. In some embodiments, radiopaque markers **640** may be cylindrical, with a bore extending therethrough along the longitudinal axis of the cylinder. The radiopaque markers **640** provided on suture rigging assembly **600** need not all have the same shape, and different shapes may be assembled to various tethers **615** to indicate the orientation of prosthetic heart valve or to identify various portions thereof. Moreover, if any of tethers **615** is improperly affixed to prosthetic heart valve or becomes improperly affixed to the prosthetic heart valve during delivery of the heart valve into the patient or during deployment, radiopaque markers **640** may help to identify which of the tethers is improperly affixed and identify its location.

[0038] Radiopaque markers **640** may be held in a fixed position on tethers **615** by lower fixture knot **630** at the distal end of the marker and by a second or upper fixture knot **645** formed in the tether at the proximal end of the marker. Fixture knots **645** and **630** capture the radiopaque marker **640** therebetween and prevent it from sliding along the length of tether **615** toward or away from attachment loop **635**. As a less preferable alternative, adhesives can be used to attach the radiopaque markers **640** at fixed positions to tethers **615**. As a result, once a radiopaque marker **640** has been identified under fluoroscopy, the user will know the position of the attachment loop **635** associated with that marker.

[0039] The use of knots to form suture rigging assembly **600** provides several advantages. Firstly, it enables adhesives to be avoided, reducing sterilization, storage and biocompatibility issues that adhesives may create. The elimination of adhesives may also reduce the formation of very small particles during the use of delivery device **100**, which particles could potentially be released into the patient's bloodstream. The use of knots throughout suture rigging assembly **600** also enables the assembly to be self-balancing, minimizing the tensile stress in any one tether **615** and increasing the overall tensile capacity of the suture rigging assembly. Finally, the various knots in each tether **615** keeps suture threads close to one another to prevent undesirable entanglement of the tethers with structures of prosthetic heart valve during deployment.

[0040] Suture rigging assembly **600** can be used to attach, load, and release a wide variety of heart valves to/from a wide variety of catheter-based delivery systems. Thus, suture rigging assembly **600** is designed to attach to a prosthetic heart valve and sustain a tensile load path between the heart valve and a delivery device as the heart valve is retracted into a sheath of the delivery device.

[0041] One way in which suture rigging assembly **600** may be used to collapse and load prosthetic heart valve V into the valve cover (e.g., valve cover **165**, **550a**, or **550b**) of delivery device **100** will now be described. Initially, suture rigging assembly **600** is attached to prosthetic heart valve V. This is accomplished by fitting some or, preferably, all of the attachment loops **635** at the distal ends of tethers **615** over respective pins on prosthetic heart valve V. Although this is described here as an

initial step, it need not be the first step in the process. Suture rigging assembly **600** may be attached to delivery device **100** first, as described below, followed by the attachment of prosthetic heart valve **V** to the suture rigging assembly **600**.

[0042] Referring to FIG. 7, a loading funnel **650** can be used to help load prosthetic heart valve **V** into valve cover. In one embodiment, loading funnel **650** may include a funnel portion **650a** located at its distal end and an elongated tubular portion **650b** located at its proximal end. Funnel portion **650a** may smoothly transition from a relatively large diameter at its distal end to a relatively small diameter where it meets tubular portion **650b**. The diameter at the distal end of funnel portion **650a** is preferably larger than the outer diameter of prosthetic heart valve **V** in its expanded condition, and the diameter of the funnel portion where it meets tubular portion **650b** is preferably about the same as the diameter of the lumen of the tubular portion **650b**. The outer diameter of tubular portion **650b** is preferably slightly smaller than the inner diameter of valve cover and the length of the tubular portion may be about the same as the length of the valve cover, such that the tubular portion can be selectively inserted into and nest within the valve cover. Funnel portion **650a** may include a plurality of slots (not shown) extending longitudinally along its inner surface. These slots are intended to accommodate the pins of prosthetic heart valve **V** and prevent them from bending laterally as the heart valve is being collapsed.

[0043] With the tubular portion **650b** of loading funnel **650** coupled to a distal end of the valve cover, controls located on the operating handle of delivery device **100** may be manipulated to cause suture catheter **655** to advance distally relative to the other components of the delivery device until the tip ring **660** of the suture catheter extends distally beyond the distal end of the tubular portion and into the interior of funnel portion **650a**. At that point, the threads of coupling ring **601** may be threaded into the threaded portion of tip ring **660** at the distal end of suture catheter **655**. FIG. 7 illustrates suture rigging assembly **600** connected to prosthetic heart valve **V** and aligned for connection to the suture catheter **655** of delivery device **100**. Suture catheter **655** may then be retracted proximally, drawing suture rigging assembly **600** and prosthetic heart valve **V** proximally along with it. As proximal movement continues, tethers **615** are drawn into the lumen of the tubular portion **650b** of loading funnel **650**. This, in combination with the sloping walls of funnel portion **650a**, causes the petals on atrial anchor of the prosthetic heart valve **V** to collapse toward the central axis of prosthetic heart valve **V** and, eventually, to enter the lumen of tubular portion **650b**. Further proximal movement of suture catheter **655** continues until the petals on ventricular anchor also collapse toward the central axis of prosthetic heart valve **V** and the prosthetic heart valve is completely collapsed and completely positioned within the lumen of the tubular portion **650b**. At that juncture, while maintaining tension on suture catheter **655**, loading funnel **650** may be removed from valve cover, leaving the fully collapsed prosthetic heart valve **V** positioned completely within the valve cover. An atraumatic tip (not shown) of the delivery device may then be coupled to the nosecone catheter and then retracted to enclose the open distal end of valve cover.

[0044] In another embodiment, the loading funnel may have a generally cylindrical shape with internal threads at one end and an internal diameter that is about the same as the inner diameter of valve cover. The internal threads may mate with external threads at the free end of valve cover to join the loading funnel to the valve cover. A smooth radius on the lumen at the free end of the funnel may help to guide the prosthetic heart valve into the funnel lumen.

[0045] Once properly loaded, delivery device **100** may be inserted into a patient and directed to a target location, such as the mitral valve annulus, at which prosthetic heart valve **V** may be deployed. To deploy prosthetic heart valve **V**, valve cover is retracted proximally over valve **V** while the valve is maintained in position by the extension catheter (or the intermediate catheter). The ventricular anchor of valve **V** will then begin to expand until only the proximal end of the valve (i.e., atrial anchor) is held in a collapsed condition by a small cup (e.g. valve retainer or can) at the distal end of extension catheter. The accurate positioning and orientation of prosthetic heart valve **V** may then be confirmed, after which suture catheter **655** may be advanced distally, relieving

tension in tethers **615** and allowing atrial anchor to escape from the valve retainer at the distal end of extension catheter and expand. Suture catheter **655** may be advanced further through the expanded prosthetic heart valve until tethers **615** slip off of the pins of the frame of the prosthetic heart valve. Suture catheter **655** may then be retracted back into outer delivery sheath, the atraumatic tip may be retracted to again close the open end of valve cover, and delivery device **100** may be removed from the patient.

[0046] As noted above, catheters such as steering catheter **410a** may incorporate tension cables to allow for deflection of the steering catheter. In some examples, steering can be provided using a single tension cable, without a corresponding return cable. In those examples, steering (which may also be referred to as catheter deflection) is performed by tensioning the single tension cable which has a distal end coupled to a distal portion of the steering catheter. After achieving deflection, there is no mechanism in these systems to actively return to a non-deflected state, rather relying on passive return to a non-deflected state. However, in other examples, including those shown and described above, steering can be provided using one or more pairs of tension cables, with one cable being tensioned to deflect the catheter, and the other cable of the pair being tensioned to actively return the catheter to a non-deflected state (or otherwise being tensioned to deflect the steering catheter in the opposite direction). In these types of systems, such as that shown in FIG. **4A**, the tension cable of a steering cable pair is typically offset by about 180 degrees from the return cable of the steering cable pair, such that each cable provides for opposite directions of deflection. Each cable of the cable pair is also typically coupled to the same actuator (e.g., knob, slider, gear) so that actuation of one of the cables of the pair simultaneously causes tensioning of one cable and relaxation of the other cable in the pair. It should also be noted that the 180 degree offset described above is not a requirement. For example, in the example shown in FIG. **4B**, the tension cable of a steering cable pair may be offset by less than 180 degrees from the return cable of the steering cable pair. Even in this situation, although the tension cable and return cable are not exactly offset diametrically, the same general principles apply regarding the tension cable and the return cable being tensioned to deflect the catheter in generally opposite directions. Further details of these types of systems are provided in U.S. Patent Application Publication No. 2023/0364387, the disclosure of which is hereby incorporated by reference herein.

[0047] Referring back to FIGS. **1** and **4A** (although this description generally applies to the example of FIG. **4B** as well), one tension cable of a steering cable pair may extend through steering catheter **410a** via lumens **421a**, and the return cable of the steering cable pair may extend through the pair of lumens diametrically opposite the lumens labeled **412a**. Similarly, a second tension cable of a second steering cable pair may extend through a third pair of the lumens, and a second return cable of the second steering cable pair may extend through a fourth pair of the lumens diametrically opposite the third pair. In some examples, only a single lumen per steering cable is used, with the proximal end of the steering cable connected to a steering actuator at the delivery device handle, and the distal end of the steering cable connected to a distal end portion of the steering catheter, for example via a steering ring. However, in other examples, including that illustrated in FIG. **4A**, a single steering cable extends distally through one lumen **412a** of the lumen pair, connects to the steering ring and loops back through the other lumen **412a** of the lumen pair. This type of configuration may provide greater strength to the steering cable compared to the single cable—single lumen example.

[0048] One issue that can occur in steering catheters that utilize cable pairs (e.g. one tension cable and one return cable per steering direction) is that either cable of the pair may have some amount of slack in the cable before being actuated. In this situation, if a user actuates a steering actuator (e.g., knob) to tension a cable of the cable pair to cause catheter deflection, but the cable that is to be tensioned includes some slack at the moment of actuation, a first amount of the actuation (e.g., an initial amount of rotation of the knob) will cause the cable to lose its slack. As the actuator causes the cable to lose its slack, the catheter does not actually begin to deflect until the slack is removed

and continued actuation of the actuation (e.g., continued rotation of the knob) applies tension to the steering cable that actually gets transmitted to the steering catheter (instead of just working to reduce the slack). The range of actuation in which tension on the cable only works to reduce slack, instead of cause the desired steering, may be referred to as a control deadband where, although the user is actuating the actuator, no steering actually results from the actuator. Stated in other words, if there is slack in the steering cable just before actuation, rotating the steering knob (or actuating the steering actuator generally) results in a non-responsive control while the slack in the steering cable is taken up. Thus, it would be preferable to reduce or eliminate the likelihood of slack that may occur in steering cables, thus reducing or eliminating the control deadband perceived by the operator during steering operations.

[0049] FIGS. **8-10** show various views of internal components of internal components of a steering assembly **1000** that may provide benefits over traditional catheter steering assemblies. Although a single steering assembly **1000** is shown and described, it should be understood that more than one of the steering assemblies may be provided with a catheter system. For example, up to three of the steering assemblies **1000** may be provided for use with delivery system **100** shown and described above, for example within handle assembly **105**, with one steering knob **110** or steering actuator corresponding to each of the three steering assemblies **1000** positioned inside of the handle assembly **105**. However, it should be understood that the concepts and features described in connection with steering assembly **1000** may be applied to any catheter system with steering capabilities, particularly those including bi-directional steering.

[0050] FIGS. **8-10** illustrate different views of an example of a steering assembly **1000**, although it should be noted that components of another steering assembly are also visible in the figures. The steering assembly **1000** may include a base **1010** which may generally provide support for other components that are attached directly or indirectly to the base **1010** to form the steering assembly **1000**. Although the base **1010** has one particular form factor in the figures, it should be understood that any other suitable form factor may be used to house and/or support the various other components of the steering assembly **1000**.

[0051] Referring now to FIG. **10**, the steering assembly **1000** may include an actuator shaft **1020** which may generally take the form of a cylindrical member. A first end of the actuator shaft may include an aperture **1022**. In one example, the aperture **1022** may be configured to receive a shaft of an actuator, such as a shaft of a steering knob **110**, such that rotation of the knob **110** in either rotational direction results in corresponding rotation of the actuator shaft **1020**. The actuator shaft **1020** may extend from the first end to a second opposite end, with a middle portion thereof extending through the base **1010**. For example, a mid-portion of the actuator shaft **1020** may be coupled to and extend through one or more supports **1024a-b**, which may include bearings or other components to help secure the actuator shaft **1020** relative to the base **1010** and to assist with the rotation of the actuator shaft **1020** relative to the base **1010**. For example, a top support **1024a** may be generally cylindrical and fixed to the actuator shaft **1020**, a bottom support **1024b** may be generally cylindrical and fixed to the actuator shaft **1020**, and a center support **1024c** may be positioned between the top and bottom supports **1024a-b**, with one or more bearings (e.g. ball bearings) being positioned between the top support **1024a** and the center support **1024c**, as well as between the bottom support **1024b** and the center support **1024c**.

[0052] Still referring to FIG. **10**, the actuator shaft **1020** may be fixed to a knob gear wheel **1026** (a portion of which is also visible in FIG. **9**), which may be a relatively small gear with teeth that mesh with teeth of a relatively large drive gear **1028**. With this configuration, rotation of an actuator (e.g. steering knob **110**) coupled to actuator shaft **1020** causes rotation of knob gear wheel **1026**, and thus rotation of drive gear **1028**, to activate a steering operation, described in further detail below.

[0053] Referring mainly to FIGS. **9-10**, drive gear **1028** may take the form of a toothed gear wheel with a central opening. The central opening of the drive gear **1028** in the illustrated embodiment

has a cross-sectional shape with two opposing flats and two rounded sections connecting the pair of flats. In other words, the central opening of the drive gear **1028** may be stadium-shaped in some embodiments, although in other embodiments any shape that permits torque translation (e.g. triangular, rectangular, etc.) may be suitable. Referring to FIGS. **10-11**, the central opening of the drive gear **1028** may receive a bottom end portion **1110** of a spool shaft **1100**. As shown in FIG. **11**, the bottom end portion **1110** of spool shaft **1100** may have a complementary shape to the central opening of the drive gear **1028**, including for example a general stadium shape with two opposing flats and rounded portions connecting the pair of flats. With this configuration, during assembly, the spool shaft **1100** may only be inserted into the drive gear **1028** in two possible orientations, either of which is acceptable and both of which result in effective torque transmission between the spool shaft **1100** and the drive gear **1028**. The bottom end portion **1110** of the spool shaft **1100** may include a through bore **1112**, although in other embodiments through bore **1112** may be omitted. If included, through bore **1112** may allow the spool shaft **1100** to be used for either a right-hand or left-hand user, for example by reversing the orientation of the spool shaft **1100** and using through bore **1112** for receiving a pin **1150** instead of using through bore **1114** (described in greater detail below).

[0054] Still referring to FIGS. **9-10**, the drive gear **1028** may include a small diameter extension extending below the gear teeth. One or more hard stop features may be provided on the drive gear **1028**. For example, a first hard stop **1028a** may be received over the small diameter extension of the drive gear **1028**, with the first hard stop **1028a** including an extension that rotates along with the drive gear **1028**. When the first hard stop **1028a** contacts another component, e.g. the smooth body of knob gear wheel **1026**, the contact will prevent further rotation of the drive gear **1028**, providing for a limit of rotation in at least one rotational direction. A second hard stop **1028b** may also be provided. For example, as shown in FIG. **9**, second hard stop **1028b** may include a hooked end. In the illustrated example, while the first hard stop **1028a** is positioned below the plane of the drive gear **1028**, the second hard stop **1028b** is positioned generally coplanar with the plane of the drive gear **1028**. However, in some embodiments, part of the second hard stop **1028b** may be coupled to the drive gear **1028** below the plane of the teeth of the drive gear **1028**, with a portion of the hard stop **1028b** (e.g. the hooked portion) being coplanar with the plane of the drive gear **1028**. With this configuration, as the drive gear **1028** is rotated in the direction of the hooked end of the second hard stop **1028b**, the hooked end will eventually intermesh with gear teeth of the knob gear wheel **1026**, binding the gears and serving as another limit of rotation. In some embodiments, the second hard stop **1028b** may serve as a rotation limiter in one rotational direction (e.g. clockwise) while the first hard stop **1028a** may serve as a rotation limiter in the other rotational direction (e.g. anti-clockwise).

[0055] Referring to FIG. **10**, two similar or identical spool assemblies are shown being coupled to the spool shaft **1100** on opposite sides of the base **1010**. As will be described in greater detail below, each spool assembly may be coupled to a steering wire (which may also be referred to as a pull-wire) of a pair of steering wires configured to provide bi-directional steering (e.g., deflection in opposite directions along the same steering direction or steering plane). Each spool assembly may include an outer spool **1200** (which may alternatively be referred to simply as a spool) and an inner spool **1300** (which may alternatively be referred to as an inner spline or a spline nut), with these components being shown in FIGS. **12** and **13**, respectively, in isolation.

[0056] FIGS. **8-10** show a first or upper spool assembly positioned on the opposite side of the base **1010** as the drive gear **1028**. FIG. **8** shows that a steering cable **1400** may extend from the outer spool **1200** where it passes into the steering catheter. For example, the steering cable **1400** may have a first portion that extends through one lumen **412a** of a lumen pair of steering catheter **410a**, distally down the steering catheter until it loops around a steering ring, and then proximally back through an adjacent lumen **412a** of the lumen pair, eventually returning to the outer spool **1200**. However, in other embodiments, as mentioned above, the steering cable **1400** may have a first end

that terminates at the outer spool **1200** and a second end that terminates at the steering ring, without the looped configuration described above. As shown in in FIG. **12**, the outer spool **1200** may include a recess in the form of a cable track **1210** which extends a distance radially inwardly toward an interior cylindrical column that generally surrounds the spline recess **1220**, described in greater detail below. This interior cylindrical column may have an outer diameter that is smaller than the outer diameter of the portions of the outer spool **1200** above and below the cable track **1210**. The steering wire **1400** may extend into the cable track **1210**, wrap at least partially around the interior cylindrical column, and terminate at a connection location **1230**. Referring in particular to FIGS. **8** and **12**, the connection location **1230** may include a recess **1232** that may generally extend downwardly to connect to cable track **1210**, so that the steering cable **1400** can be routed from the cable track **1210**, up the recess **1232**, and then into an enlarged area **1234** of the connection location **1230**. The connection location **1230** may include a through bore **1236** in the enlarged area **1234**, the through bore **1236** configured to receive a fastener such as a screw **1238**, shown in FIG. **8**. The fastener **1238** may be fixedly coupled to the outer spool **1200** via the through bore **1236**, and an end of the steering cable **1400** may be coupled to the outer spool **1200** via the fastener **1238**. For example, the end(s) of the steering cable **1400** may be received in the enlarged area **1234** and clamped down via a head of the screw or fastener **1238** that is engaged with the through bore **1236**. With this configuration, as will be described in greater detail below, as the outer spool **1200** rotates, the fixed coupling between the steering cable **1400** and the fastener **1238** ensures that the steering cable **1400** can be tensioned while the cable track **1210** guides the portions of the steering cable **1400** that are being drawn into the outer spool **1200** and wound around the inner cylinder of the outer spool **1200**.

[0057] Each spool assembly, including the upper spool assembly best shown in FIGS. **8-10**, may also include an inner spool **1300** received within the spline recess **1220** of the outer spool **1200**. Referring to FIG. **13**, each inner spool **1300** may include a relatively thin bottom flange **1310** having a relatively large diameter, that transitions to a center column section which may include a smooth base **1320** having a smaller diameter than the flange **1310**, and a spline section **1330** extending upwardly from the smooth base **1320**. Although the steering cable is generally configured to wrap around the track **1210** of the outer spool **1200**, a portion of the innermost loop of the steering cable may come into contact with smooth base **1320**. In some examples, the spline section **1330** is taller than both the smooth base **1320** and the flange **1310**. The spline section **1330** may include a plurality of splines, which may also be referred to as alternating peaks or ridges and valleys or recesses which all extend in a longitudinal direction (e.g., parallel to the axis of rotation). The number of splines in the spline section **1330** may be more than or fewer than shown in FIG. **13**, and the number may be increased or decreased to provide more or less (respectively) resolution of pre-tension on the steering cable **1400**, which is described in greater detail below. Notably, the spline section **1330** preferably has a number and configuration of splines that is complementary to the shape of the radially-inner surface of the spline recess **1220** of the outer spool so that each ridge of the spline section **1330** may be received within a corresponding valley of the spline recess **1220**, and each ridge of the spline recess **1220** may be received within a corresponding valley of the spline section **1330**. The inner spool **1300** may also include a central longitudinal recess **1340** that has a stadium-shape or another torque-transmitting shape that is complementary to the shapes of the bottom end portion **1110** and the top end portion **1120** of the spool shaft **1100**.

[0058] Referring to FIGS. **8-11**, the spool shaft **1100** may include a cylindrical central portion **1130** between the stadium-shaped top end portion **1120** and bottom portion **1110**, and as best shown in FIG. **10**, the central portion **130** may be received within a cylindrical through bore of the base **1010**, such that the top end portion **1120** protrudes above the base **1010** and the bottom end portion **1110** protrudes below the base **1010**. In this configuration, one inner spool **1300** may be slid over the top end portion **1120** of the spool shaft **1100**, and another inner spool **1300** may be slid over the bottom end portion **1110** of the spool shaft **1100**, with the two inner spools **1300** both having their

flanges **1310** in contact with the base **1010**. At this point, one outer spool **1200** may be slid over the top inner spool **1300**, and another outer spool **1200** may be slid over the bottom inner spool **1300**. When the inner spool **1300** is received within the outer spool **1200**, the splines of the spline section **1330** engage the corresponding splines of the spline recess **1220**, so that the inner spool **1300** is rotationally locked to the outer spool **1200**. Referring to FIGS. **10-11**, the spool shaft **1100** may include a central aperture **1131** that can align with a lumen of the base **1010** (as shown in FIG. **10**). During assembly, a pin or other device may be temporarily inserted through the lumen of the base and into the central aperture **1131** to assist with locking rotation of the spool shaft **1100** during assembly. After assembly is completed, the pin may be removed from the lumen and the central aperture **1131** to again allow for rotation of the spool shaft **1100** relative to the base **1010**.

[0059] After the outer spools **1200** are received over the inner spools **1300**, and the inner spools **1300** are received over the spool shaft **1100**, coupling members may be used to lock the axial positions of the spool assemblies to the spool shaft **1100**. For example, referring to FIG. **11**, a through bore **1114** may be provided in the bottom end section **1110** of the spool shaft **1100**, and another through bore **1124** may be provided in the top end section **1120** of the spool shaft. As best shown in FIGS. **8-9**, these through bores **1114**, **1124** may be configured to be positioned just beyond the top and bottom ends of the spool assemblies, respectively, so that one pin **1150** or a similar device (e.g. a cotter pin) can be passed through each of the through bores **1114**, **1124**, ensuring the spool assemblies are axially locked between the pin **1150** and the base **1010**. At this point, the drive gear **1028** may be positioned over the bottom end portion **1110** of the spool shaft **1100** (which may be longer than the top end portion **1120** so that there is enough space to accommodate the drive gear **1028**), and assembly of the steering assembly **1000** may be substantially complete once the drive gear **1028** is fixed to the spool shaft **1100** and intermeshes with the knob gear wheel **1026**. As noted above, the steering assembly **1000** may be encased within a housing which the actuator shaft **1020** protrude from so that an actuator, such as steering knob **110**, may be fixed to the actuator shaft **1020**.

[0060] Although FIG. **8** only shows the top spool assembly with the steering cable **1400** being coupled to the outer spool **1200** via fastener **1238** and extending toward the steering catheter, it should be understood that the bottom spool assembly may have an identical configuration except that the steering cable coupled to the bottom spool assembly may be wound about the spool in an opposite direction compared to the top spool assembly. Further, it should be understood that the steering cable coupled to the bottom spool assembly may extend toward the steering catheter and into a pair of lumens (e.g. a pair of lumens **412a**) which may be substantially diametrically opposed to the pair of lumens through which the top steering cable **1400** extends. However, as explained above, in some examples, particularly in those with three or more pairs of steering cable lumens, the two opposing tension/return cables corresponding to a single bi-directional steering direction need not be exactly diametrically opposed, and rather may be slightly offset from being diametrically opposed.

[0061] With the configuration described above, when actuator shaft **1020** is rotated in a first direction (e.g., by rotating steering knob **110** in the first direction), the knob gear wheel **1026** rotates in the first direction, causing the drive gear **1028** to rotate in the opposite direction. As the drive gear **1028** rotates, it causes the spool shaft **1100** to rotate due to the corresponding torque-transmitting shapes of center opening in the drive gear **1028** and the bottom end portion **1110** of the spool shaft **1100**. As the spool shaft **1100** rotates, it causes both inner spools **1300** to rotate in the same direction, again due to the corresponding torque-transmitting shapes of the top and bottom end portions **1120**, **1110** of the spool shaft **1100** and that of the longitudinal recess **1340** of the inner spool **1300**. As the inner spool **1300** rotates, the engagement between the splines of the spline section **1330** of the inner spool **1300** with the interior splines of the spline recess **1220** causes the outer spool **1200** to rotate along with the inner spool **1300**. As the outer spools rotate **1200** in the same direction as each other, one spool will tend to tension its connected steering cable as the

steering cable winds around the spool, while the other spool will tend to slacken its connected steering cable as the steering cable unwinds from the spool, because as noted above, the steering cables are wound in opposite directions along the two spools. Thus, actuating the actuator shaft **1020** (e.g. by rotating steering knob **110**) in a first direction will tension one steering cable of the pair and simultaneously slacken the other steering cable of the pair to cause bending of the steering catheter in a first direction, while actuating the actuator shaft **1020** in a second direction opposite the first direction will have the opposite effect, causing bending of the steering catheter in a second direction opposite the first direction. Thus, bidirectional steering may be achieved with a single steering knob **110**. Referring briefly to FIG. **10**, each outer spool **1200** may include a gasket **G** that generally overlies a part of the relevant steering cable so that, as slack is introduced into the steering cable, the steering cable generally remains within the track **1210** of the outer spool **1200**. [0062] As explained above, one benefit of the configuration shown and described in connection with FIGS. **8-13** is that a single steering actuator may be manipulated to cause simultaneous tensioning (by winding around one spool) of one steering cable of a pair and slackening (by unwinding around another spool) of the other steering cable of a pair to achieve bidirectional steering with a single steering wheel (or knob, slider, etc.).

[0063] However, another benefit of this configuration is that it allows for substantial or complete elimination of a perceived control deadband due to slack existing in steering cables just prior to actuating the steering mechanism. For example, during assembly of the outer spool **1200** (to which the end(s) of the steering cable is fixed), the outer spool **1200** can be situated over the inner spool **1300** in a plurality of different rotational orientations. For example, referring to FIGS. **12-13**, if the spline section **1330** includes twenty splines (and the spline recess **1220** includes a corresponding twenty splines), there are twenty unique rotational positions that the inner spool **1300** may have relative to the outer spool **1200**, with each rotational position changing in increments of about eighteen degrees. It should be understood that the example of twenty splines is merely exemplary, and more splines may be included to provide even more unique rotational positions (with smaller degrees of offset between different positions) or fewer splines may be included to provide for fewer unique rotational positions (with larger degrees of offset between different positions). With this configuration, when the outer spool **1200** is being slipped over the inner spool **1300** to form the spool assembly, the user may choose between these various unique rotational positions, which may allow the user to fine tune how much tension is on the steering cable at the time of assembly, which may be referred to as pre-tensioning. Thus, pre-tension on the steering cables may be set during assembly, with the desired amount of pre-tension being determined by which incremental rotational position the outer spool **1200** has relative to the inner spool **1300** at assembly. This can be performed for both spool assemblies, so that both steering cables have a desired amount of pre-tension. By pre-tensioning the steering cables, neither steering cable will have slack (or meaningful slack) when the knob **110** (or other steering actuator) is actuated to rotate the actuator shaft **1020** to deflect the steering catheter in the desired direction, and thus any perceived control deadband will be fully (or substantially fully) eliminated. Further, because actuation of the actuator shaft **1020** causes simultaneously winding of one steering cable around its spool and unwinding of the other steering cable around its spool, slack is not likely to be introduced into either steering cable at any time during the procedure.

[0064] Although the steering assemblies herein are described in general terms of a steering catheter for delivering a prosthetic heart valve, it should be understood that these concepts and/or features may be implemented in any steering catheter without needing significant alteration of the components. Further, although the steering assemblies are described in connection with bi-directional steering (e.g., with two spool assemblies controlling the tension and return directions), at least some of the benefits described herein may be achieved with unidirectional steering (e.g., with one spool assembly only controlling only a single direction of steering).

[0065] Although the invention herein has been described with reference to particular embodiments,

it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A catheter system comprising: a steering catheter having a wall and a first lumen formed within the wall; a first steering cable extending through the first lumen along a length of the wall of the steering catheter toward a distal end portion of the steering catheter, the first steering cable being coupled to the distal end portion of the steering catheter; and a steering assembly including a first spool assembly, the first spool assembly including a first outer spool, a first inner spool received at least partially within a recess of the first outer spool, and a spool shaft having a first end portion received at least partially within a recess of the first inner spool, wherein the first end portion of the spool shaft and the recess of the first inner spool are each shaped so that rotation of the spool shaft about a longitudinal axis thereof transmits torque to the first inner spool to cause rotation of the first inner spool about the longitudinal axis of the spool shaft; wherein the first inner spool includes a spline section forming a plurality of splines, and the recess of the first outer spool has a shape that is complementary to the plurality of splines, such that when the first inner spool is received at least partially within the recess of the first outer spool, the plurality of splines engage the complementary shape of the recess of the first outer spool to prevent rotation of the first inner spool relative to the first outer spool about the longitudinal axis of the spool shaft; and wherein the first steering cable is fixed to the first outer spool and configured to be wound around the first spool assembly upon rotation of the first outer spool about the longitudinal axis.
2. The catheter system of claim 1, wherein first inner spool is configured to be received within the first outer spool in any one of a plurality of unique relative rotational positions, so that the first steering cable is configured to have a first amount of pretension in a first one of the plurality of unique relative rotational positions and a second amount of pretension in a second one of the plurality of unique relative rotational positions, the first amount of pretension being different than the second amount of pretension.
3. The catheter system of claim 1, wherein the steering assembly includes a base, the spool shaft has a central portion received within the base, and the first spool assembly is positioned on a first surface of the base.
4. The catheter system of claim 3, wherein the steering assembly includes a second spool assembly positioned on a second surface of the base opposite the first surface of the base.
5. The catheter system of claim 4, wherein the second spool assembly includes a second outer spool, and a second inner spool received at least partially within a recess of the second outer spool.
6. The catheter system of claim 5, wherein the first inner spool is identical to the second inner spool, and the first outer spool is identical to the second outer spool.
7. The catheter system of claim 5, wherein the spool shaft has a second end portion received at least partially within a recess of the second inner spool, and the second end portion of the spool shaft and the recess of the second inner spool are each shaped so that rotation of the spool shaft about the longitudinal axis thereof transmits torque to the second inner spool to cause rotation of the second inner spool about the longitudinal axis of the spool shaft.
8. The catheter system of claim 7, wherein the second inner spool includes a spline section forming a plurality of splines, and the recess of the second outer spool has a shape that is complementary to the plurality of splines of the spline section of the second inner spool, such that when the second inner spool is received at least partially within the recess of the second outer spool, the plurality of splines of the spline section of the second inner spool engage the complementary shape of the recess of the second outer spool to prevent rotation of the second inner spool relative to the second

outer spool about the longitudinal axis of the spool shaft.

9. The catheter system of claim 8, further comprising: a second lumen formed within the wall of the steering catheter; and a second steering cable extending through the second lumen along a length of the wall of the steering catheter toward the distal end portion of the steering catheter, the second steering cable being coupled to the distal end portion of the steering catheter, wherein the second steering cable is fixed to the second outer spool and configured to be wound around the second spool assembly upon rotation of the second outer spool about the longitudinal axis.

10. The catheter system of claim 9, wherein the first steering cable is fixed to the first outer spool and the second steering cable is fixed to the second outer spool such that, when the spool shaft is rotated about the longitudinal axis to cause simultaneous rotation of the first outer spool and the second outer spool, the first steering cable winds around the first spool assembly while the second steering cable unwinds from the second spool assembly.

11. The catheter system of claim 10, wherein the first lumen is positioned on a generally opposite side of the steering catheter compared to the second lumen.

12. The catheter system of claim 11, wherein tensioning the first steering cable is configured to deflect the distal end portion of the steering catheter in a first steering direction, and tensioning the second steering cable is configured to deflect the distal end portion of the steering catheter in a second steering direction, the first steering direction being generally opposite the second steering direction.

13. The catheter system of claim 10, wherein the first lumen is positioned diametrically opposite to the second lumen.

14. The catheter system of claim 13, wherein tensioning the first steering cable is configured to deflect the distal end portion of the steering catheter in a first steering direction, and tensioning the second steering cable is configured to deflect the distal end portion of the steering catheter in a second steering direction, the first steering direction and the second steering direction being positioned in a same plane but in opposite directions along the same plane.

15. The catheter system of claim 10, further comprising a drive gear, the drive gear having a central opening, the second end portion of the spool shaft being received within the central opening, wherein the second end portion of the spool shaft and the central opening of the drive gear are each shaped so that rotation of the drive gear about the longitudinal axis of the spool shaft transmits torque to the spool shaft to cause rotation of the spool shaft about the longitudinal axis of the spool shaft.

16. The catheter system of claim 15, wherein the steering assembly further comprises: an actuator shaft having a first end portion and a second end portion, the second end portion being received within and coupled to the base; and a gear wheel having a central opening, the first end portion of the actuator shaft being received within the central opening of the gear wheel such that rotation of the actuator shaft about a longitudinal axis thereof causes rotation of the gear wheel about the longitudinal axis of the actuator shaft, wherein the gear wheel is operably coupled to the drive gear such that rotation of the gear wheel transmits torque to the drive gear.

17. The catheter system of claim 16, wherein the steering assembly is received within a handle of a delivery system, and the actuator shaft has a terminal end protruding outside of the handle.

18. The catheter system of claim 17, further comprising a steering knob coupled to the terminal end of the actuator shaft such that rotation of the steering knob causes corresponding rotation of the actuator shaft.

19. The catheter system of claim 16, wherein the steering assembly includes a first hard stop operably coupled to the drive gear, the first hard stop configured to prevent rotation of the drive gear beyond a first threshold amount of rotation in a first rotational direction.

20. The catheter system of claim 19, wherein the steering assembly includes a second hard stop operably coupled to the drive gear, the second hard stop configured to prevent rotation of the drive

gear beyond a second threshold amount of rotation in a second rotational direction opposite the first rotational direction.
